PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

COURSE OUTLINE FOR
B.VOC (FOOD PROCESSING & QUALITY MANAGEMENT)
AS PER NEP 2020

W.E.F SESSION 2024-25

COURSE OUTLINE FOR B.VOC
(FOOD PROCESSING & QUALITY MANAGEMENT)
AS PER NEP 2020 W.E.F. SESSION 2024-25

SEMESTER-I										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 101	Core Discipline	Food Biochemistry	2	2	4	40	10	40	10	100
FPQM 102	Core Discipline	Dairy Technology	2	2	4	40	10	40	10	100
FPQM 103	Core Discipline	Food Quality Control	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
		Total Credits			24					600

SEMESTER-II										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 201	Core Discipline	Food Packaging-1	2	2	4	40	10	40	10	100
FPQM 202	Core Discipline	Principles of Food Processing and Preservation	2	2	4	40	10	40	10	100
FPQM 203	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
		Total Credits			24					600

Job Role: Dairy and Food Processor

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in B.Voc. Food Processing & Quality Management = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

PROGRAMME OUTCOMES

PO1. Students know about the scientific method to food science problems.PO2. Know about quantitative reasoning skills to food science data.

PO3. Apply critical thinking and analytical evaluation to contemporary food scienceinformation and literature.

PO4. Apply principles from general chemistry, biology, physics, stastics and mathematicsto food science problems.

SEMESTER I

SEMESTER I

COURSE TITLE		INTRODUCTION TO BIOCHEMISTRY
1.	**Course Code**	FPQM-101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration / LP (Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical.(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical foreach unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80 TH / PR + 20 IA) i. The exam of this course will be of 3 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Dairy & Food Processor
7.	**Course Objective**	Study of various nutrients and maintaining the biochemical properties of food products.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 Understand the role of different nutrients in various food products. CO2 Know the sources, classification and properties of various nutrients in food products. CO3 Study the structure of various types of nutrients.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i. The syllabus of this paper has been divided into FOURunits. ii. Examiner will set a total of NINE questions comprising two questions from each unit, includingQuestion No. 1 (compulsory) of short answer typecovering the whole syllabus. iii. The students are required to attempt one question fromeach unit and the entire Compulsory Question No. 1. iv.All questions carry equal marks.

11. Course Content

UNIT-I

General introduction and importance of Biochemistry. Different nutrients of food, Carbohydrates - introduction, sources, classification, basic structure, biochemical functionsand metabolic pathways of carbohydrates
{Glycolysis and Kreb cycle)

UNIT-II

Proteins - Introduction, classification, sources, structural organisations and biochemical functions. Amino acids and their importance (Essential and Non-Essential Amino acids) and properties of
amino acids.

UNIT-III

Lipids - Introduction, classification, sources and biochemical functions. Brief introduction to fattyacids, essential fatty acids, their physical and
chemical properties. Oxidation and Rancidity.

UNIT-IV

Vitamins - Introduction, classification, sources, biochemical functions, daily requirements & deficiency diseases of fat and water soluble vitamins. Minerals - Introduction, classification as major and minor elements, sources, biochemical functions, daily requirements & deficiency diseases
of Ca, Fe, I, P, Na, K, F & Zn.

Practical Work:

1. Preparation of standard solutions of acid and alkali.
2. Determination of pH using indicators and with pHmeter.
3. Determination of moisture in a given sample.
4. Determination of ash in a given sample.
5. Determination of acidity in a given sample.
6. Estimation of fat by soxhlet method.
7. Estimation of free fatty acid of given sample.

12. Suggested Resources forReading

1. Principles of Biochemistry by Lehinger AL, 1995,CBS Publisher, New Delhi.
2. Text Book of Biochemistry by Rama Rao AVSS,1999, LK and S Publishers, Visakhapatanm, Indian.
3. Laboratory Manual in Biochemistry by Jayaraman J,
   1996, New Age International Ltd., New Delhi

Note: Latest edition of text books may be used.

	COURSE TITLE	**DAIRY TECHNOLOGY**
1.	**Course Code**	FPQM-102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration /LP (Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical. (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks andTime**	100 Marks (80 TH / PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculatorswill not be provided by the University/College to the examinees
6.	**Job Role**	Dairy & Food Process
7.	**Course Objective**	To supervise manufacture of dairy products like milk, butter, yogurt, cheese and their preservation.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 To develop knowledge among students about various aspects of dairy industry. CO2 To study quality standards and production of various types of milk and milk products. CO3 To study the role of dairy farming in Indian economy
9.	**Pedagogy of theCourse Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters andcandidates**	i) The syllabus of this paper has been divided into FOURunits. ii) Examiner will set a total of NINE questions comprising two questions from each unit, includingQuestion No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Milk: Definition, composition, physicochemical properties of milk and its constituents, nutritional value, overview of processes used in milk and milk products- pasteurization, homogenization and standardization, dye detection test, platform tests, milk borne diseases. Microbiology of milk, types of spoilage and its preventions. Types of market milk: Toned milk, Double Toned milk, Standardized Milk, Recombined milk, reconstituted milk, fortified milk. Effluent treatment of dairy plant disposal.

UNIT-II

Technology underlying manufacture of Milk products: Butter, cream, butteroil, cheese, yoghurt,paneer, chhana, ice-cream

UNIT-III

Composition, standards, manufacturing: Process, equipments and defects during manufacturing and storage of dairy products and by products (paneer, dahi, milk powder-skimmed milk and whole milkpowder, casein, whey concentrate, lactose, ghee residue).

UNIT-IV

Dairy development in India: - Present status, future prospective and its role in Indian economy. Important government initiative. Role of dairy development organizations (NDRI, Amul) in dairy
development

Practical Work:

1. Sampling of milk and milk products formicrobiological analysis
2. Platform test for milk analysis.
3. To determine fat in milk by gerber method.
4. To determine specific gravity of given sample of milk by lactometer.
5. Visit and study a nearby milk union/ dairy and prepare a checklist of problems in procurement and milk distribution.
6. Detection of various adulterants in milk.
7. Preparation of sterilized flavored milk.
8. To prepare pasteurized milk.
9. Methylene blue reduction test in milk

12. Suggested Resources forReading

1. Fluid milk industry, J.S Handerson, A.V.I PublishingCompany, USA1
2. Indian Dairy products, K.T. Acharya Publication
3. Milk Hygiene in milk production processing anddistribution, F.A.O Publication.
4. Sukumar.1983.Outlines of Dairy Technology.De, Oxford University Press.

Note: Latest edition of text books may be used.

	COURSE TITLE	FOOD QUALITY CONTROL
1.	**Course Code**	FPQM-103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration /LP (Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical.(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical foreach unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks andTime**	100 Marks (80 TH / PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Dairy & Food Processor
7.	**Course Objective**	To determine and establish quality standards for food products and Is responsible for ensuring that food products produced meet standards set by both the organization and regulatory authorities.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 To understand the different principles and functions of food quality control department. CO2 To understand various food laws and regulations CO3 Understand the role of food quality assurance CO4 Understand the principles of total quality management systems & standards in India. CO5 Understand the principles of total quality management.
9	**Pedagogy of theCourse Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters andcandidates**	i) The syllabus of this paper has been divided into FOURunits. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer typecovering the whole syllabus. iii) The students are required to attempt one question fromeach unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Objectives, functions and principles of quality control. Difference between food quality controland quality assurance, assessment of raw materials and finished products (quality control) of baking industry.

UNIT-II

Total quality management (TQM) – Principles of quality management, good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control. Quality Attributes: Size, shape, colour, aroma and texture. Microbial quality control: an overview

UNIT-III

Food laws and regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO 9000 series etc, Food standards and safety Act: salient provisions and prospects, role of various food standards in India- PFA, FPO, AGMARK and BIS.

UNIT-IV

Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control. Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituentsof bakery products.

PRACTICAL WORK:

1. Techniques of quality assessment of different naturaland processed foods.
2. Documentation of details of baking ingredients,process and finished products used in baking industry.
3. Quality evaluation of milk and milk products
4. Sensory methods for measuring food quality assessment of raw materials used and processed bakery products
5. Instrumental methods for measuring food quality assessment of raw materials used and processed bakery products
6. Study of cleaning and sanitizers used in pre and post- operative processes in bakery industry
   Listing of quality control agencies at national and international levels

12. Suggested Resources forReading

1. InteazAlli (2003). Food Quality Assurance Principlesand Practices. CRC Press.
2. J Andres Vasconcellos (2003) Quality Assurance forthe Food Industry. A Practical Approach. CRC Press.

Note: Latest edition of text books may be used.

SEMESTER-I

	COURSE TITLE	FOOD PACKAGING
1.	**Course Code**	FPQM-201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration / LP (Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical.(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for eachunit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80 TH / PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will notbe provided by the University/College to the examinees
6.	**Job Role**	Dairy& Food Processor
7.	**Course Objective**	Monitors packaging of various foods and their subsequentCompatibility and handles all categories of packaging such as primary, secondary and tertiary packaging for food products.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 To enable the students to understand about packagingand packaging materials, compatibility of various food items with packaging materials
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT-I

Packaging Technology: Definitions, Functions of packaging, Properties of packaging material in relation to their functions, package design, Tests on packaging materials, Gas and water vapor transmission rates, types of containers- primary&secondary, flexible & rigid, hermetic & non-hermetic.

UNIT-II

Packaging materials: Flexible films and pouches, Plastics used and their Specific applications. Rigid and Semi rigid plastic packaging methods. Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card Board packaging

UNIT-III

Metal Cans: properties of metals for packaging material, Types of Metallic Cans - Tin cans and Aluminum cans, steel plate and functions of its various constituents, Importance of Open top sanitary cans, Lacqueringtypes and applications, three piece cans and two piece cans. Introduction to Canning operations – Can Reformer, Flanger, Seaming, Can closures, Sterilization of jars and bottles.

UNIT-IV

Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of Flexible plastic containers, Aseptic packaging, shrink packaging, gas packaging, vacuum and modified atmospherepackaging – principles, applications. Estimation of shelf life, analysis of storage requirement, accelerated storage Suggested Resources for Reading:

Practical Work:

1. To determine thickness of paper and paper board.
2. To identify different types of packaging materials.
3. Demonstration.of measurement of cartons' dimensionsas per organizational standards.
4. Demonstration of measurement of dimensions of bottlemouth, cans and their caps.
5. Demonstration of sealing processes used in foodindustry.
6. Demonstration of filling process.

12. Suggested Resources forReading

1. Coles, R., Dowell, D.M., Kirwan, J. "Food Packaging Technology". Black Well Publishing
2. Ltd.ISBN084939788X, 2003.
3. NIIR Board: Food Packaging Technology Handbook National Institute of Industrial Research, New Delhi (2004) Pirenger O.G.and A.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH, Germany (2000).

Note: Latest edition of text books may be used.

COURSE TITLE		PRINCIPLES OF FOOD PROCESSING AND PRESERVATION
1.	**Course Code**	FPQM-202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration / LP(Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical.(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical foreach unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80 TH / PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Dairy & Food Processor
7.	**Course Objective**	Working on different methods of processing of different foods and their preservation techniques.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 To understand the basic principles & objectives of food processing. CO2 To study different means of food preservation and their subsequent utilization.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOURunits. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question fromeach unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Introduction- Historical developments of food preservation, Principles of food preservation, benefits of food preservation. Food Spoilage and to study its causes (microbial, physical or chemical)

UNIT-II

Preservation by heat: Heat resistance ofmicroorganisms, thermal death curve, heattreatments and their effects on food- Boiling, steam under pressure, blanching, pasteurization, canning, aseptic packaging, cans and container types, spoilage of canned foods, heat penetration in cans.

UNIT-III

Preservation by low temperature- refrigeration storage, requirements of refrigerated storage, changes of foods during refrigerated storage, refrigeration load, Freezing and frozen storage- Slow and quick freezing, factors determining freezing rate, freezing methods, changes in foods during freezing, frozen food storage, Freezing curve, Intermediate moisture foods- their advantages and problems. Drying, dehydration and concentration- Types of drying, types of dryers,food concentration and their methods, changes in foods during dehydration and concentration.

UNIT-IV

Microwave Heating- Properties, mechanism and its food applications. Preservation by radiations Ultraviolet and ionizing radiations, its applications in foods. Chemical Preservation: Types, uses and effects of class I & class II preservatives in foods.

Practical Work:

1. Study of blanching process in vegetables.
2. Dehydration of foods.
3. Preservation of food products by low temperature.
4. Use of chemicals in preservation of food.
5. Microwave cooking of food.
6. Cut out examination of canned foods.
7. Visit to food industry.

12. Suggested Resources forReading

1. Food Science by Potter NN, 2006, CBS Publishers,New Delhi.
2. Food Microbiology by Frazier WC & West Hoff DC, 1998, Tata Macgraw Hill Publishers, New Delhi.
3. 3. The Technology of food preservation by Desrosier and Desrosier.
4. Fellows, P. (2000) Food Processing Technology, Principles & Practices 2nd Edition Woodhead, Cambridge.

Note: Latest edition of text books may be used.

	COURSE TITLE	FOOD MICROBIOLOGY
1	**Course Code**	FPQM-203
2	**Typology of Course**	Core Discipline
3	**Course Duration / LP(Lectures) (Practical)**	30 hours of lectures for Theory + 6 0 hours of practical.(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical foreach unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH / PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Dairy & Food Processor
7	**Course Objective**	To identify various microorganisms associated with food andto analysis the microbiological quality of different food products.
8	**Course Outcomes**	On successful completion of the course, the students will be able to CO1 To understand the isolation methodology of microorganisms CO2 To identify the microorganisms of food products of plant and animal origin. CO3 To learn about Food borne diseases and microorganisms.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOURunits. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 Course Content

UNIT-I

Introduction : Discovery of microbial world, application of microbiology and its relevance to food technology, theory of spontaneous generation, germ theory of disease, kosh's postulates, pure culture concept, nature and properties of prokaryoticand eukaryotic.

UNIT-II

Microscopy- light microscope- resolving power, magnification, bright field, dark field, electronmicroscopy, scanning electron microscope

UNIT-III

Food as nutrients for various micro-organism, factors affecting the growth and survival of microorganisms in food, general features and importance of different group of bacteria, yeast and molds important in food, methods of microbial examination of food and food products. Definitionof growth, growth cycle, growth rate, generation time, measurement of growth.

UNIT-IV

Food spoilage- microbial and biochemical aspects offood spoilage, role of bacteria, yeast and molds in food spoilage, spoilage of cereals and cereal products, fruits and vegetables, meat and meat products, milk and milk products, food borneillness, food intoxication and food infection, bacterial food poisoning by streptococcus aureus, clostridium botulinum, salmonella, E. Coli, Clostridium perfringes, Listeria monocytogenes, Campylobacter jejuni

PRACTICAL WORK:

1) To study different parts of microscope.
2) Study the instruments (autoclave, hot air oven, incubation, laminar flow, pH meter, and spectrophotometer) of microbiological laboratory.
3) Sterilization and disinfection of equipments used in food microbiology laboratory.
4) To determine the no. of microorganism withhaemocytometer.
5) Isolation of fungi from food material.
6) Preparation of media, slant and broths required in microbial analysis of food.
7) Study of incubation tests of heated canned food.
8) Micro biological analysis of egg, cereals productsand food products.
9) Determination of TPC in food products.
10) Determination of Coliform counts.
11) Microbiological analysis of water-samples.

12 Suggested Resources forReading

1. Microbiology by Pelczar M.J., Chan E.C.S & Krieg N.R., McGraw Hill Co.
2. General Microbiology by Stanier R.Y., In graham J.L., Wheelies M.L. & Painter P.R. Macmillan press Ltd., London
3. Laboratory Manual in Microbiology by GunaSeKaran P, New age international Ltd., New Delhi.
4. Frazier WC and Westoff DC "Food Microbiology" Tata macro Hill Publishing
5. Adams MR and Moss MO "Food Microbiology"New age international Ltd,

Note: Latest edition of text books may be used.